1.7 Equilibria

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.7 Equilibria
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

At 450℃ the value for K_c for the below reaction is 60.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

The equilibrium moles of H_2 and I_2 are 2 mol and 0.3 mol, respectively.

How many moles of HI (g) is present at equilibrium?

A. 6

B. 0.001

C. 36

D. 0.1

[1 mark]

Question 2

Ethyne and hydrogen are formed from methane, and a dynamic equilibrium is established.

$$2CH_4(g) \rightleftharpoons 3H_2 + C_2H_2(g)$$

What are the units for K_c?

A. mol dm^{-3}

B. mol^2 dm^{-6}

C. mol^3 dm^{-9}

D. mol^4 dm^{-12}

[1 mark]

Question 3

Upon heating, dinitrogen tetroxide will dissociate into nitrogen dioxide.

$$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$$

At a particular temperature, the equilibrium partial pressure of nitrogen dioxide was 0.67 atm and dinitrogen tetroxide 0.33 atm.

What is the numerical value of K_p at this temperature?

A. 1.36

B. 0.49

C. 0.65

D. 2.03

[1 mark]

Question 4

Nitrogen monoxide and oxygen can be formed from the thermal decomposition of nitrogen dioxide.

$$2NO_2\,(g) \rightleftharpoons 2NO\,(g) + O_2\,(g)$$

In an experiment 4 mol of nitrogen dioxide were put into a 1 dm^3 container and heated to a constant temperature. The equilibrium mixture contains 0.8 mol of oxygen.

What is the value of the equilibrium constant, K_c, at the temperature of the experiment?

A. 0.8^2 x 0.8 / 4^2

B. 1.6 x 0.8 / 2.4^2

C. 1.6^2 x 0.8 / 2.4

D. 1.6^2 x 0.8 / 2.4^2

[1 mark]

Question 5

The Contact process is used for the industrial production of sulfuric acid. The three main chemical reactions involved in the contact process are listed below.

$$S\,(s) + O_2\,(g) \rightarrow SO_2\,(g)$$

$$SO_2\,(g) + \frac{1}{2}\,O_2\,(g) \rightarrow SO_3\,(g)$$

$$SO_3\,(g) + H_2O\,(l) \rightarrow H_2SO_4\,(l)$$

Which statement about this process is correct?

A. In the first stage, a large excess of air under high pressure is used to improve the yield.

B. Two of the three stages are equilibria.

C. All three stages are exothermic.

D. In the final stage, SO_3 is absorbed by water droplets.

[1 mark]

Question 6

The dissociation of gas X_2Y_4 into XY_2 is represented in the equation below.

$$X_2Y_4\,(g) \rightleftharpoons 2XY_2\,(g) \quad \Delta H^\theta = +60\ \text{kJ mol}^{-1}$$

At constant pressure, if the temperature of the equilibrium mixture of the gases is increased, will the volume of the mixture increase or decrease and why?

A. The volume will increase, but only because of a shift of equilibrium towards the right.

B. The volume will increase, both because of a shift of equilibrium towards the right and also because of thermal expansion.

C. The volume will stay the same because any thermal expansion could be exactly counteracted by a shift of equilibrium towards the left.

D. The volume will decrease because a shift of equilibrium towards the left would more than counteract any thermal expansion.

[1 mark]

Question 7

The table gives the concentrations and pH values of the aqueous solutions of two compounds, P and Q. Either compound could be an acid or a base.

	P	**Q**
Concentration	2 mol dm^{-3}	2 mol dm^{-3}
pH	9	6

Student X concluded that Q is a strong acid.

Student Y concluded that the extent of dissociation is lower in Q (aq) than in P (aq).

Which of the students are correct?

A. X only

B. Y only

C. Both X and Y

D. Neither X nor Y

[1 mark]

Question 8

A compound, WX, will dissociate into an X^- ion and a W^+ ion in solution. This is represented by the equation below.

$$WX(aq) \rightleftharpoons X^-(aq) + W^+(aq)$$

2 dm^3 of a 0.4 mol solution of WX was heated to a constant temperature until equilibrium was established. The equilibrium mixture is found to contain 0.1 mol of the W^+ ion.

What is the correct expression for the equilibrium constant K_c.

A. $0.1^2 / 0.3$

B. $0.15 / 0.05^2$

C. $0.05^2 / 0.15$

D. $0.05 / 0.15$

[1 mark]

Question 9

Ammonia is produced from hydrogen and nitrogen in the Haber process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

A mixture of 2.00 mol of nitrogen, 3.00 mol of hydrogen and 1.98 mol of ammonia is allowed to reach equilibrium in a sealed container with a volume of 1 dm^3 under certain conditions. It was found that 1.64 mol of ammonia were present in the equilibrium mixture.

What is the value of K_c under these conditions?

A. 0.10

B. 0.03

C. 0.22

D. 2.17

[1 mark]

Question 10

When 5.00 mol of A reacts with 5.00 moles of B in a sealed vessel of volume 1 dm^3 and total pressure of 1.5 atm an equilibrium is established which contains 3.00 moles of C.

$$A + B \rightleftharpoons 2C$$

What is the equilibrium partial pressure of A?

A. 2.00 atm

B. 0.525 atm

C. 0.350 atm

D. 0.450 atm

[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- **Write out the equation for *Kc***
  - $K_c = \dfrac{[HI\,(g)]^2}{[H_2\,(g)]\,[I_2\,(g)]}$
- **Rearrange the equation for *Kc* making HI the subject**
  - $[HI\,(g)] = \sqrt{K_c\,[H_2\,(g)]\,[I_2\,(g)]}$
- **Substitution into the equation**
  - $[HI\,(g)] = \sqrt{60 \times (2) \times (0.3)} = 6$

B is incorrect as the K_c equation has been rearranged incorrectly. You may have come to this answer by making your initial expression for *K*c the wrong way around like below. Remember its products over reactants, i.e. $K_c = \dfrac{[H_2\,(g)]\,[I_2\,(g)]}{[HI\,(g)]^2}$.

C is incorrect as the HI was not squared. Remember, the stoichiometric coefficient (the big number in front of the species in the equation) must turn into an exponent (the number that is raised to the power of).

D is incorrect as both the K_c equation has been rearranged incorrectly (as explained in A), and the HI was not squared (as explained in C).

2

The correct answer is **B** because:

- The numerator (top bit) in the K_c equation is raised to the power of 4. The denominator is raised to the power of 2.
- $K_c = \dfrac{[H_2\,(g)]^3\,[C_2H_2\,(g)]}{[CH_4\,(g)]^2}$
- **Maths tip:** When you are dividing indices you simply minus them.
- 4 - 2 = 2, therefore the numerator units becomes $(mol\ dm^{-3})^2 = mol^2\ dm^{-6}$.

A, C & D are incorrect as none of these units fit with the correct expression of K_c.

3

The correct answer is **A** because:

- **Write out the equation for K_p**
  - $K_p = \frac{p(NO_2)^2}{p(N_2O_4)}$
- **Substitution into the equation**
  - $K_p = \frac{(0.67)^2}{(0.33)} = 1.36$

B & **C** are incorrect as K_p must be greater than 1 as the equilibrium partial pressure of the products is greater than the reactants.

D is incorrect as the value for the NO_2 (g) equilibrium partial pressure has not been squared. Remember the stoichiometric coefficient (the big number in the equation) becomes the exponent (the little number raised to the power of) in the K_p equation.

4

The correct answer is **D** because:

- Create an initial, change and equilibrium table. Remember to use **the molar ratio of the change.** We do not have to worry about dividing the equilibrium moles by the volume here as the volume is 1 dm^3. (2/1= 2, 3/1 = 3 etc)

	$2NO_2$ (g) $\rightleftharpoons$	$2NO$ (g) +	O_2 (g)
Initial moles	4	0	0
Change	-1.6	+1.6	+0.8
Equilibrium moles	2.4	1.6	0.8

- **Write out the equation for *Kc***

- $K_c = \dfrac{[NO\ (g)]^2\ [O_2\ (g)]}{[NO_2\ (g)]}$

- **Input the values calculated in the equilibrium moles row of the table above**
    - $K_c = 1.6^2 \times 0.8 / 2.4^2$

A is incorrect as the incorrect value for NO (g) has been used. The molar ratio of the change has not been used to calculate the equilibrium moles.

B is incorrect as the value for the NO (g) equilibrium moles has not been squared. Remember the stoichiometric coefficient (the big number in the equation) becomes the exponent (the little number raised to the power of) in the K_c equation.

C is incorrect as The value for NO_2 (g) equilibrium moles has not been squared.

5

The correct answer is **C** because:

- Every step in the contact process releases heat, making them exothermic.
- The first step is combustion making it the most exothermic of the three steps.
- The second step is an exothermic oxidation reaction.
- The third step is so exothermic that it actually causes the water to evaporate. The reaction stated is a simplified version of the actual process.

A is incorrect as at first glance, this statement looks correct! However, an excess of air is not in fact used to improve the yield. It is used so that the sulphur dioxide produced is already mixed with oxygen for the next stage.

B is incorrect as only the second step is a reversible reaction and therefore can be deemed an 'equilibria'.

D is incorrect as the third step is so exothermic that it actually causes the water to evaporate. The reaction stated is a simplified version of the actual process. The SO_3, in reality, is first dissolved in a small quantity of concentrated sulphuric acid first to prevent this from happening.

6

The correct answer is **B** because:

- The forward reaction is endothermic, as the sign of ΔH is positive

- The second step is an exothermic oxidation reaction.
- The third step is so exothermic that it actually causes the water to evaporate. The reaction stated is a simplified version of the actual process.

A is incorrect as at first glance, this statement looks correct! However, an excess of air is not in fact used to improve the yield. It is used so that the sulphur dioxide produced is already mixed with oxygen for the next stage.

B is incorrect as only the second step is a reversible reaction and therefore can be deemed an 'equilibria'.

D is incorrect as the third step is so exothermic that it actually causes the water to evaporate. The reaction stated is a simplified version of the actual process. The SO_3, in reality, is first dissolved in a small quantity of concentrated sulphuric acid first to prevent this from happening.

6

The correct answer is **B** because:

- The forward reaction is endothermic, as the sign of ΔH is positive
  - Therefore, an increase in temperature would shift the equilibrium to the right-hand side of the reaction (to oppose the change) thus increasing the amount of XY_2 (g).
- You can see that there are more moles of gas on the right-hand side of the reaction.
  - As moles of a gas are proportional to volume, the volume will increase.
- When you increase the temperature of any gas, you are giving the gases more kinetic energy, and therefore the gas molecules move further apart.
  - This will result in an increased volume, which is thermal expansion.

A is incorrect as thermal expansion is a thermodynamic law that cannot be ignored. When you increase the temperature of any gas, you are giving the gases more kinetic energy, and therefore the gas molecules move further apart. This will result in an increased volume.

C & **D** are incorrect as the equilibrium shifts to the right, not the left. Remember the forward reaction is endothermic, so an increase in temperature would shift it to the right.

7

The correct answer is **B** because:

- We can see from the question that both solutions have the same concentration.
- The question also states that the solutions can be acidic or alkaline.
- An acidic concentration contains more hydrogen ions (H^+) and has a pH lower than 7.
- An alkaline solution contains more hydroxide ions (OH^-) and has a pH higher than 7.
- The pH in solution **P** is further away from neutral (either higher or lower) so therefore has a greater dissociation.

A & **C** are incorrect as they both state that student **X** is correct, **Q** is not a strong acid as it has a pH close to neutral.

D is incorrect as student **Y** is correct.

8

The correct answer is **C** :

- Create an initial, change and equilibrium table. Remember to use **the molar ratio of the change.**
- Remember to divide the equilibrium moles by 2 (volume is 2 dm^{-3}) to get the equilibrium concentrations.

	$WX (aq) \rightleftharpoons$	$X^- (aq) +$	$W^+ (aq)$
Initial moles	0.4	0	0
Change	-0.1	+0.1	+0.1
Equilibrium moles	0.3	0.1	0.1
Equilibrium conc	0.15	0.05	0.05

- **Write out the equation for *Kc***
    - $K_c = \dfrac{[X]^- [W]^+}{[WX]}$

- **Input the values calculated in the equilibrium concentration row from the table above**
    - $K_c = \frac{(0.05)^2}{(0.15)}$

A is incorrect as the equilibrium moles have been used and not the equilibrium concentrations.

B is incorrect as the K_c equation and thus the values are the wrong way around: [reactants] / [products] instead of [products] / [reactants].

D is incorrect as the equilibrium concentration for products has not been squared, which must be done because there were the same moles of X^- and W^+ at the start of the reaction (zero) therefore, X^- and W^+ exist in a 1:1 molar ratio is this equation.

9

The correct answer is **B** because:

- Here the equilibrium has shifted to the left as the moles of ammonia has decreased.

	N_2 (g) +	$3H_2$ (g) $\rightleftharpoons$	$2NH_3$ (g)
Initial moles	2	3	1.98
Change	+0.17	+0.51	-0.34
Equilibrium moles	2.17	3.51	1.64

- By quickly calculating the moles present at equilibrium, you can see that when inserting the values into the K_c equation for this reaction, your answer will be less than 1 as the numerator (top bit) is less than the denominator (bottom bit).
    - $K_c = \frac{[NH_3]^2}{[N_2]\,[H_2]^3}$
    - $K_c = \frac{(1.64)^2}{(2.17)\,(3.51)^3}$
    - $K_c = 0.02866$

10

The correct answer is **B** because:

- Create an initial, change and equilibrium table. Remember to use **the molar ratio of**